DSA-301 Runbook

What is left to do, in order. Run everything from the folder holding `editor.py`.

Already done, so not repeated here: the venv is installed, and the five `.md` files are generated in `--link-only` form. `LC200` is already pushed and verified as the worked example. Full explanations live in `README.md` / `DSA301-Editor-Guide.pdf`.

1. Start Chrome - every session, and leave it open

```
"%LOCALAPPDATA%\Google\Chrome\Application\chrome.exe" --remote-debugging-port=9222 --user-data-dir="%USE
    RPROFILE%\chrome-debug-profile"
```

Log in to HackerRank if it asks. The separate `--user-data-dir` is **not optional** - Chrome 136 and later refuse the debugging port on your normal profile, so your everyday shortcut will not work.

Chrome must stay open for the whole of step 2; `editor.py` drives that window.

2. Push, one file at a time

```
venv\Scripts\python.exe editor.py DSA301_LEC01-10.md --apply --tests --samples 3
venv\Scripts\python.exe editor.py DSA301_LEC11-20.md --apply --tests --samples 3
venv\Scripts\python.exe editor.py DSA301_LEC21-30.md --apply --tests --samples 3
venv\Scripts\python.exe editor.py DSA301_LEC31-40.md --apply --tests --samples 3
venv\Scripts\python.exe editor.py DSA301_LEC41-50.md --apply --tests --samples 3
```

Order does not matter and re-running is safe - matching is by title and every edit happens in place, so a re-run never creates a duplicate.

`--samples 3` matters here. These files are `--link-only`, so there is no input-format prose and the visible sample cases are the **only** statement of the stdin contract. Two examples is thin; three is safer.

Budget about **11 seconds per challenge** - roughly 45 minutes for all five, Chrome open throughout.

What a good challenge looks like

```
id=592083   Classwork: LC200 Number of Islands
   differs: problem_statement, input_format, constraints, output_format
   saved and verified
   test cases replaced: 8
   samples: 2 marked as sample
```

`saved and verified` is the line that counts: `editor.py` re-fetched the challenge from the REST API and every field it sent came back matching.

Two things to watch for

`not matched : N` at the top of a file. Those are skipped and nothing is written for them. It means either no live challenge carries that title, or **two** do - about 18 problems match a duplicate pair. `editor.py` refuses to guess and prints both IDs. Pick the canonical one and target it directly:

```
venv\Scripts\python.exe editor.py DSA301_LEC21-30.md --only "<part of the title>" --apply --tests --samp
    les 3
```

`--only` matches titles **in the file you name**, so the substring and the file have to agree. `DSA301_LEC01-10.md --only "LC94"` matches nothing and exits 1, because LC94 lives in `DSA301_LEC11-20.md`.

`test cases APPENDED: 5 -> 10`. The upload added to the old cases instead of replacing them. It is counted a failure, but the 5 stale cases are still attached and have to be removed by hand.

If something breaks

symptom	fix
`Cannot reach Chrome on 127.0.0.1:9222`	Chrome is not on the debug port, or is on your normal profile. Redo step 1
`FAILED verification: <fields>`	the save did not take. Re-run that one challenge with `--only`. Do not leave a statement field empty - HackerRank refuses a save that blanks them
`SKIPPED: found N editors, expected 4`	the edit page did not present its four CodeMirror boxes. Re-run that challenge; `editor.py` skips rather than risk writing into the wrong boxes
`HTTP 403` on a test-case upload	CSRF token rejected. Re-run - the token is read fresh from the live page
students see no samples	re-run that challenge with `--tests`, and read the `samples:` line

Two limits, before you go looking for a control that does not exist: **challenges cannot be deleted** - the admin panel has no destructive control, so editing in place is the only option. And **slugs are immutable** - a full field rewrite leaves the URL alone.

If you change a problem later

Then and only then do you regenerate. Edit the Python, never the markdown - `make_md.py` overwrites all five `.md` files on every run.

```
venv\Scripts\python.exe make_md.py --link-only
venv\Scripts\python.exe editor.py <the file> --only "<part of the title>" --apply --tests --samples 3
```

Problems live in `make_md.py` (LEC 01-30), `dp_problems.py` (LEC 31-40), `graph_problems.py` (LEC 41-50), and their source links in `problem_links.py`. Add a `SANITY` entry with an answer you worked out by hand - `make_md.py` writes nothing if a reference solution disagrees with one. README section 4 has the detail, including the other two generation modes.